YUQIANG LIN

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LinkedIn ◇ Personal Website ◇ GitHub ◇ Google Scholar

SUMMARY

PhD researcher in computer vision, multimodal large language models, and agentic AI. Research spans model training, video understanding, and efficient inference, with experience deploying AI systems on edge devices.

EDUCATION

University of Bath Bath, United Kingdom
Integrated PhD in Advanced Automotive Propulsion Systems Sep 2022 – Oct 2027(expected)
Research focus: Computer vision and multimodal AI for intelligent transportation systems.

Chongqing University of Technology Chongqing, China
Bachelor of Engineering in Vehicle Engineering, GPA:89/100, Rank:1/82 Sep 2018 – Jun 2022

EXPERIENCE

AI Research Intern, LEC AI Sep 2026 – Nov 2026 *London, UK*
- Benchmarked video understanding methods to support an agentic pipeline for automated content creation.

Visiting Researcher, University of Washington Sep 2025 – Dec 2025 *Seattle, USA*
- Collaborated with UW researchers on vision-language models for traffic anomaly understanding.

PUBLICATIONS

[ECCV Workshop 2026, first author] TAU-Agent: An Agentic Retrieval-Augmented Framework for Traffic Anomaly Understanding. [paper link]
- Developed an agentic RAG framework for traffic anomaly video question answering, using video captioning and open-vocabulary tracking to retrieve visual evidence.
- Achieved 2nd place on the NVIDIA AI City Challenge 2026 Track 3 in-domain benchmark.

[ECCV Workshop 2026, first author] TAU-R1: Visual Language Model for Traffic Anomaly Understanding. [paper link]
- Developed decomposed-QA SFT and task-specific GRPO, raising 4-class AP by 50 percent and LLM-Score from 3.02 to 4.66 over base Qwen3-VL, outperforming evaluated open-source and commercial baselines.
- Collected, annotated, and released *Roundabout-TAU*, a roadside video benchmark for traffic anomaly understanding.
- Deployed a two-layer VLM framework on NVIDIA Jetson AGX Orin, achieving 2.1× real-time processing speed.

[IEEE T-ITS, under review, first author] Edge-Assisted Multi-Camera Vehicle Tracking Framework for Real-Time and Scalable Deployment. [paper link]
- Proposed an edge–server framework integrating detection, tracking, vehicle re-identification, and cross-camera association for real-time urban traffic perception.
- Optimised edge inference and workload management to sustain real-time throughput.
- Engineered a distributed perception pipeline for reproducible deployment and scalable data management.

[Nature Scientific Data, under review, first author] RoundaboutHD: High-Resolution Real-World Urban Environment Benchmark for Multi-Camera Vehicle Tracking. [paper link]
- Collected, annotated, and released *RoundaboutHD*, a real-world benchmark for multi-camera vehicle tracking, detection, single-camera tracking, geo-mapping and re-identification.

Additional Publications: ICMV 2024, ICPAMI 2024 (first author); 2 SAE 2022 papers (co-author).

SKILLS

Languages: Python(7 years), C/C++(2 years), MATLAB(3 years)
Tools & Frameworks: PyTorch, OpenCV, Hugging Face Transformers, TRL, Linux, Docker, Git, TensorRT

SELECTED AWARDS

- **UKRI EPSRC AAPS CDT Scholarship**, University of Bath – fully funded PhD.
- **National Scholarship of China**, Chongqing University of Technology – top 0.2%.